

ARE YOU RED TEAM READY?

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CONTENTS

CONTEXT

- Vulnerability Assessment concept;
- Penetration testing concept;
- Red team concept;

FRAMEWORK

- Traditional cycle;
- Red team cycle;
- Red team realistic simulation;
- Answering different questions;

CASE STUDY

- Hacking scenario;
- Hacking milestones;
- Find targets;
- Get physical access;
- Compromise target;
- Exfiltrate secret;
- Do not get caught!;

CHALLENGES

- Assembling the team;
- Execution;
- To be careful of;
- To do as a priority;

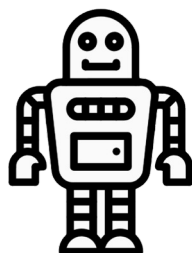
NEXT

- Future of red teams;
- Can you name those devices?
- Answers

VULNERABILITY ASSESSMENT CONCEPT



LARGE SCALE



AUTOMATED



CHEAP



REGULAR

PRENETHRATION TESTING CONCEPT



FOCUSED



SKILLED AND MORE MANUAL



COSTS MORE



ON DEMAND/LESS REGULAR

RED TEAM CONCEPT



SCENARIO BASED



HIGHLY SKILLED

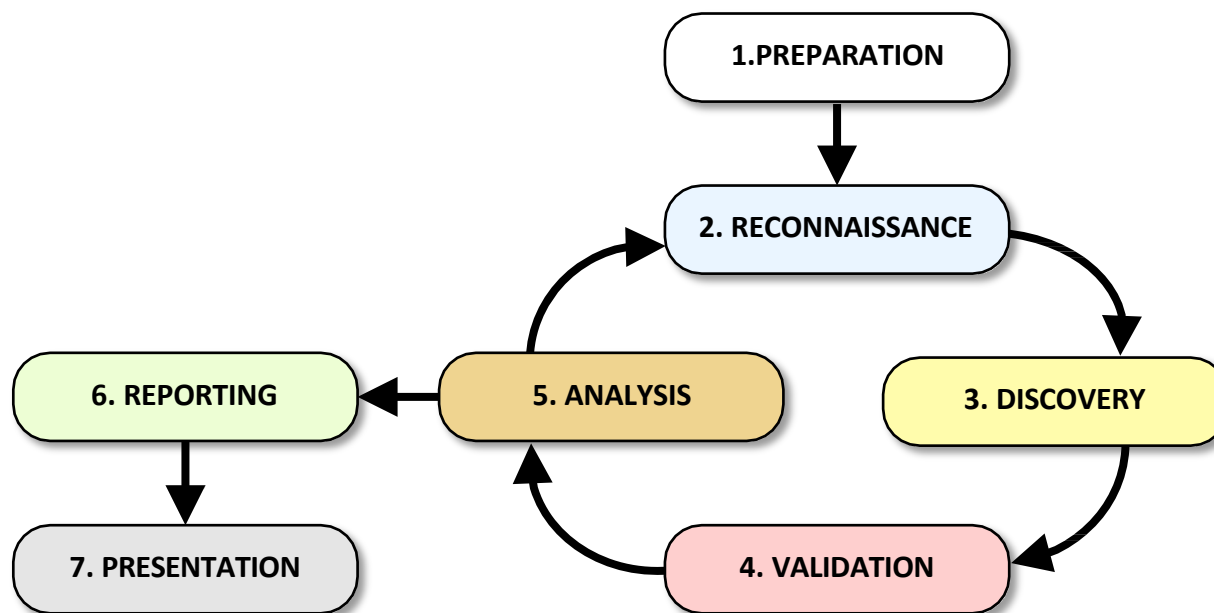


EXPANSIVE

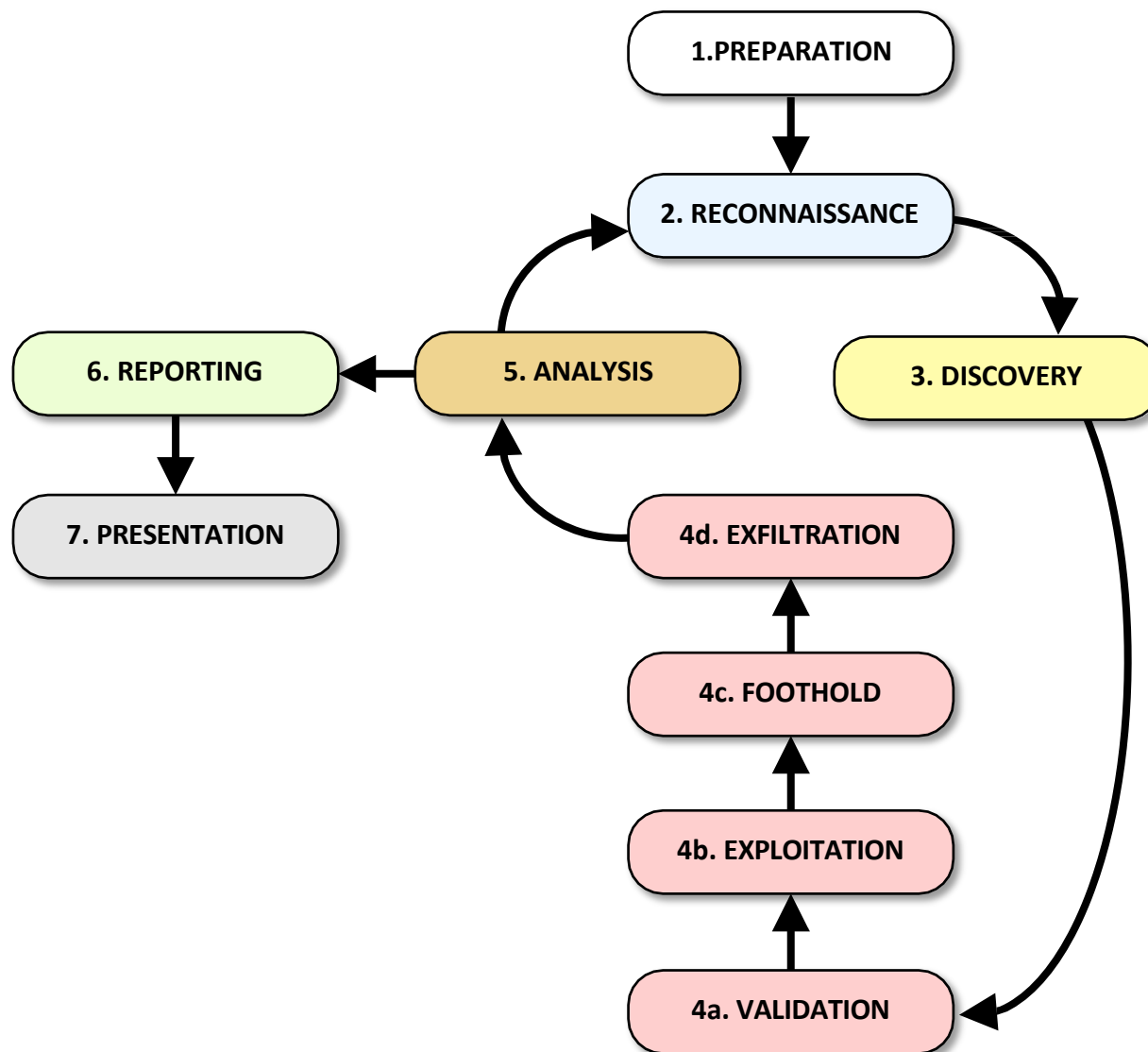


ON DEMAND/WHEN NEEDED

TRADITIONAL CYCLE



RED TEAM CYCLE

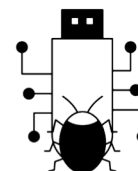
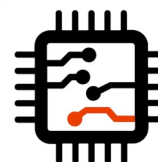


RED TEAM REALISTIC SIMULATION

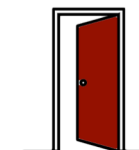
PHYSICAL / LOGICAL / SOCIAL



EMULATE HACKING TECHNICS



ESTABLISH PERSISTENCE



EXTRACT DATA



DEMONSTRATE



ANSWERING DIFFERENT QUESTIONS!

VULNERABILITY ASSESSMENT

DO WE HAVE VULNERABILITIES?

PENETRATION TESTING

CAN MY WEBSITE BE HACKED?

RED TEAM EXERCISE

CAN MY CONTRACTS BE STOLEN?

HACKING SCENARIO

OBJECTIVES

- FINANCIAL CONTRACT GOT STOLEN 2 MONTHS AGO;
- NOW STORED IN SECURED SERVER IN SECURED ROOM;
- **CAN THE CONTRACT BE STOLEN AGAIN?**

INFORMATION GIVEN

- NAME OF THE CONTRACT;
- 3x INDIVIDUALS' NAME RELEVANT TO THE CONTRACT;
- THE COUNTRY WHERE THE CONTRACT IS LOCATED.

HACKING MILESTONES

1. FIND TARGETS

2. GET PHYSICAL ACCESS

3. COMPROMISE ASSET

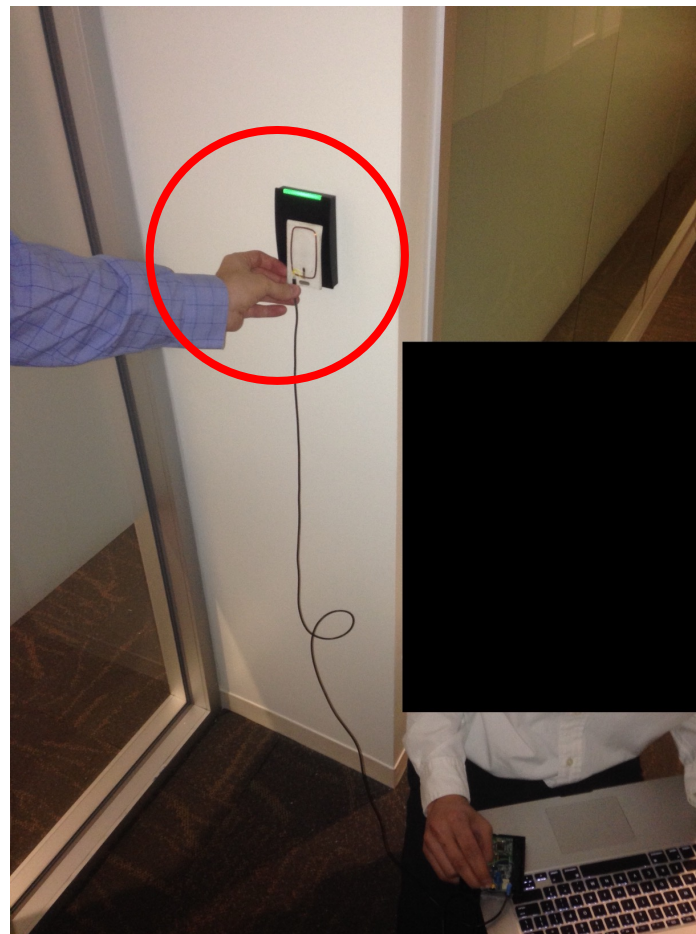
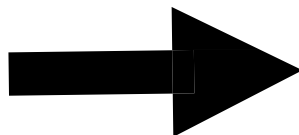
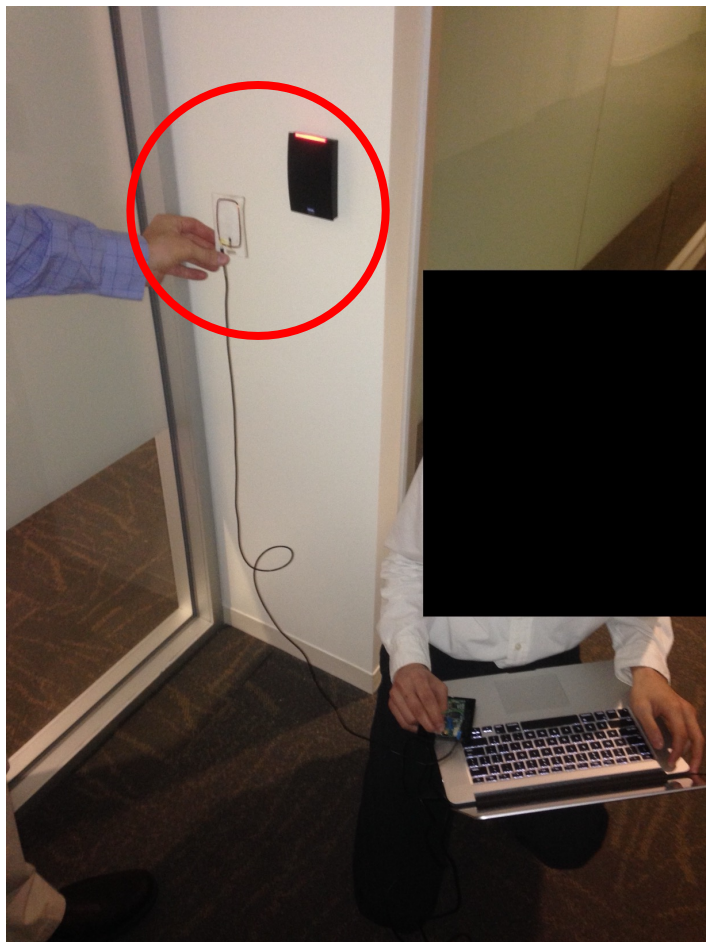
4. EXFILTRATE SECRET

5. DO NOT GET CAUGHT!

FIND TARGETS



GET PHYSICAL ACCESS







DO NOT GET CAUGHT!



ASSEMBLING THE TEAM





PROFESSIONALISM



MANDATE

TO BE CAREFUL OF

RENAMING PEN TEST TO RED TEAM

KEEPING STAFF HAPPY

TRAINING

STAY SAFE AND DON'T BECOME A TROJAN HORSE

HOW REALISTIC CAN YOU BE

SCENARIOS TOO WIDE/AMBITIOUS

TO DO AS A PRIORITY

CLEAR MILESTONES

STAKEHOLDERS ENGAGEMENT

BE REALISTIC

PROVIDE COMPREHENSIVE THREAT VIEW

GET A TEAM WITH DIFFERENT SKILLS

FUTURE OF RED TEAMS

MORE DEMANDS

PURPLE TEAM

OFFENSIVE LEGITIMISATION

CAN YOU NAME THOSE DEVICES?



ANSWERS

Ref	Device Name	Device Purpose
1	BusPirate	Intercept and analyse electric signals to convert into data
2	Cas Interface 3	Program Conditional Access Module) CAM Cards
3	Lock Picking Kit	Opening locks
4	Pocket Squirrel	Intercept ETHERNET Traffic
5	ODBDII Connector	Read car's data
6	Salae Logic 16	Intercept and analyse electric signals to convert into data
7	Proxmark III	Snigging, reading and cloning RFID
8	Bash Bunny	Emulate USB Devices
9	DTV Card	Receive Airplane Transponder Signal
10	Lan Turtle	Emulate LAN Ethernet Adapter - Intercept ETHERNET Traffic
11	Pineapple MarkV	Intercept WIFI
12	Rubber Ducky	Simulate keyboard for payload
13	High Gain Antenna	Reach far range WIFI Network
14	SSD to SATA PCB	Recover data from damaged HD
15	Shikra	Intercept and analyse electric signals to convert into data
16	Ubertooth One	Bluetooth Interception
17	VGA logger	Intercept VGA Video Signal
18	PS2 and USB Key Logger	Intercept keyboard strokes
19	Long Range WIFI Booster	Access long range WIFI Network
20	USB Wifi Dongle	Extra Wifi card for rogue access point
21	USB Wifi Dongle	Extra Wifi card for rogue access point
22	USB Stick with Ramdumpfs	Dump a Computer Memory
23	HD SATA PCB	Recover data from damaged HD
24	FaceDancer21	USB Fuzzer
25	HackRF One	Intercept Radio Signals
26	Pineapple MarkIV	Intercept WIFI
27	Throwing Star LAN Tap	Passive Network Tap
28	Pineapple MarkIV	Intercept WIFI
29	JTAG Connector	Connect to debug serial PINs from hardware
30	High Gain Antenna	Reach far range WIFI Network

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